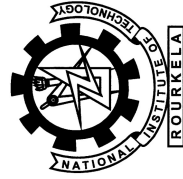


Buckling Performance Evaluation of Functionally Graded Graphene-Reinforced Composite Plates under Thermal-Mechanical Loads

Rohit Acharya



Department of Civil Engineering
National Institute of Technology Rourkela

Buckling Performance Evaluation of Functionally Graded Graphene-Reinforced Composite Plates under Thermal-Mechanical Loads

*Thesis submitted in fulfillment
of the requirements for the degree of*

Bachelor of Technology

in

Civil Engineering

by

Rohit Acharya

(Roll Number: 121CE1129)

based on research carried out

under the supervision of

Prof. Shishir Kumar Sahu



December, 2024

Department of Civil Engineering
National Institute of Technology Rourkela



Department of Civil Engineering
National Institute of Technology Rourkela

Prof. Shishir Kumar Sahu
Professor HAG

December 2, 2024

Supervisor's Certificate

This is to certify that the work presented in the thesis entitled *Buckling Performance Evaluation of Functionally Graded Graphene-Reinforced Composite Plates under Thermal-Mechanical Loads* submitted by *Rohit Acharya*, Roll Number 121CE1129, is a record of original research carried out by him under my supervision and guidance in partial fulfillment of the requirements of the degree of *Bachelor of Technology in Civil Engineering*. Neither this thesis nor any part of it has been submitted earlier for any degree or diploma to any institute or university in India or abroad.

Shishir Kumar Sahu

Acknowledgment

First and foremost, I would like to express my heartfelt gratitude to Prof. Shishir Kumar Sahu, Department of Civil Engineering, National Institute of Technology Rourkela, for his unwavering guidance, support, and encouragement throughout this research work. His deep knowledge, practical insights, and patient mentorship helped shape this project and pushed me to explore beyond the surface.

This research journey has been a learning experience beyond textbooks and simulations. Prof. Sahu's constant motivation and willingness to discuss every small idea or doubt I had truly made a difference. Working under his supervision has helped me grow not just as a student but as a researcher preparing for international academic opportunities.

I am also thankful to NIT Rourkela for offering me a platform to carry out this research, along with access to labs, simulation tools, and an academic environment that allowed me to think freely and dive deep into my field of interest.

Finally, I extend my thanks to my family and friends for their belief in me throughout this phase. Their emotional support has been my silent strength in aiming higher.

Abstract

This study presents a computational framework to analyze the buckling performance of functionally graded graphene-reinforced composite (FG-GRC) plates under coupled thermal and mechanical loading conditions. The research integrates advanced material modeling with finite element simulation to address structural stability challenges faced in high-performance civil and mechanical systems. The composite material properties are derived using an extended Halpin-Tsai formulation, accounting for graphene volume fraction gradients across the plate thickness. Buckling behavior is evaluated through linear eigenvalue analysis using ABAQUS, where critical load responses are studied across varying aspect ratios, boundary conditions, temperature fields, and graphene distributions.

The results demonstrate a clear enhancement in structural stability with increased graphene reinforcement, particularly in thermally sensitive environments. Functionally graded designs are shown to outperform uniformly distributed composites by enabling better stress distribution and higher buckling loads. This approach opens up design opportunities in infrastructure systems requiring durability, lightness, and thermal resilience—particularly in environments prone to complex stress conditions and vibrational fatigue.

The study's methodology and findings offer a foundational base for further experimental validation, material optimization, and practical deployment in applications demanding long-term stability under hybrid loads.

Keywords: Functionally Graded Composites, Graphene-Reinforced Plates, Buckling Analysis, Finite Element Method, Thermal-Mechanical Loads, Halpin-Tsai Theory, ABAQUS Simulation, Structural Stability, Material Optimization.

Contents

Supervisor’s Certificate	ii
Acknowledgment	iii
Abstract	iv
List of Figures	viii
List of Tables	x
1 Introduction	1
1.1 Background of Structural Failures and Thin Plate Stability	1
1.2 Emergence and Need for Functionally Graded Materials	2
1.3 Mechanical and Thermal Behavior of Graphene Nanoplatelets	2
1.4 Functionally Graded Graphene-Reinforced Composites (FG-GRCs)	4
1.5 Buckling – Theory, Significance, and Risk	4
1.5.1 Critical Buckling Load	5
1.5.2 Thermo-Mechanical Buckling	5
1.5.3 Significance and Risk Mitigation	5
1.6 Research Problem and Knowledge Gaps	7
1.7 Research Objectives	8
1.8 Scope and Assumptions	9
1.8.1 Scope of the Study	9
1.8.2 Assumptions	9
1.9 Objectives of the Study	10
1.9.1 Specific Objectives	10
2 Methodology	11
2.1 Material Modeling of FG-GRC	11
2.1.1 2.1 Composition Description	11
2.1.2 Gradient Index Profile	12
2.1.3 Effective Young’s Modulus via Halpin–Tsai Model	13
2.1.4 Density and Poisson’s Ratio	14

2.2	Governing Equations and Buckling Formulation	15
2.3	Loading and Boundary Conditions	16
2.3.1	Boundary Conditions	16
2.3.2	Loading Types	16
2.4	Meshing and Convergence	17
2.4.1	Element Selection	17
2.4.2	Mesh Convergence Criteria	17
2.4.3	Mesh Sizes Tested	17
2.5	Simulation Tools	18
2.6	Outlook	19
3	Experimental Results	20
3.1	Simulation Setup and Parameters	20
3.1.1	Geometric Configuration	20
3.1.2	Boundary Conditions and Loading	20
3.1.3	Material Properties	21
3.1.4	Mesh Configuration and Convergence	21
3.2	Summary of Simulation Cases	22
3.3	Critical Buckling Load Results	22
3.3.1	Effect of Gradient Index	22
3.3.2	Effect of Boundary Conditions	23
3.3.3	Effect of Aspect Ratio (a/h)	23
3.3.4	Effect of Plate Width-to-Length Ratio (a/b)	24
3.3.5	Effect of Maximum Graphene Volume Fraction	25
3.3.6	Effect of Plate Thickness	26
3.3.7	Effect of Number of Graphene Layers	27
3.3.8	Effect of Temperature	28
3.3.9	Effect of Reinforcement Percentage	29
3.3.10	Combined Thermal–Mechanical Buckling	30
3.3.11	Imperfection Sensitivity Analysis	31
3.3.12	Mesh Sensitivity Analysis	32
3.4	Buckling Mode Shape Visualization	34
3.4.1	Post-Buckling Load–Displacement Behavior	35
3.5	Comparison of Buckling Loads Across Configurations	36
3.5.1	Summary Table of Buckling Loads	37
4	Discussion	38
4.1	Influence of Gradient Index	38
4.2	Boundary Conditions and Their Impacts	38
4.3	Aspect Ratio and Plate Dimensions	38

4.4	Volume Fraction and Graphene Layers	39
4.5	Thermal Effects	39
4.6	Combined Thermo-Mechanical Loading	39
4.7	Imperfection Sensitivity	39
4.8	Mesh Independence and Accuracy	40
4.9	Post-Buckling Response	40
4.10	Comparison of Configurations	40
4.11	Design Implications and Future Applications	40
5	Conclusion	41
5.1	Key Findings	41
5.2	Conclusion	42
5.3	Future Scope	42
	References	43

List of Figures

1.1	(a) Different graphene-based configurations derived from sheet rolling and stacking; (b) Atomic structure and bonding in graphene nanoplatelets (GNPs).	3
1.2	Comparison of buckling modes in FG-GRC plates under different load conditions.	6
2.1	Schematic diagram showing the graded graphene volume fraction across the plate thickness, with applied thermal and compressive loads under SSSS boundary conditions.	12
2.2	Variation of graphene volume fraction $V_g(z)$ through the thickness of the plate for different values of gradient index n .	13
2.3	Variation of effective Young's modulus $E(z)$ through the plate thickness for different gradient indices n using the Halpin–Tsai model. The modulus increases with graphene content, demonstrating stronger reinforcement towards the surface.	14
2.4	Hypothetical variation of $\rho(z)$ and $\nu(z)$ across the plate thickness for different values of gradient index n .	15
3.1	Variation of Critical Buckling Load with Aspect Ratio (a/h)	24
3.2	Critical Buckling Load vs. Plate Width-to-Length Ratio (a/b)	25
3.3	Critical Buckling Load vs. Maximum Graphene Volume Fraction (V_{g0})	26
3.4	Critical buckling load vs. Plate Thickness	27
3.5	Critical Buckling Load vs. Number of Graphene Layers	28
3.6	Variation of Critical Buckling Load with Temperature under SSSS condition	29
3.7	Variation of critical buckling load with graphene reinforcement percentage	30
3.8	Combined Thermal–Mechanical Buckling Behavior	31
3.9	Effect of Geometric Imperfections on Buckling Capacity	32
3.10	Effect of Mesh Size on Critical Buckling Load	33
3.11	Buckling mode shape of FG-GRC plate under SSSS condition, Gradient Index $n = 2$	34
3.12	Buckling mode shape of FG-GRC plate, CCCC condition, Gradient Index $n = 1$	34
3.13	Buckling mode, SSCC condition with combined thermal-mechanical loading	35

3.14 Post-buckling-load-displacement curve for SSSS and CCCC boundary conditions	36
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List of Tables

3.1	Summary of Simulation Cases with Gradient Indices and Loading Conditions	22
3.2	Critical buckling load (kN) vs. Gradient index under SSSS condition	22
3.3	Critical buckling loads (kN) for $n = 2$ under different boundary conditions .	23
3.4	Critical buckling load (kN) for varying aspect ratios under SSSS condition .	23
3.5	Critical Buckling Load (kN) vs. Plate Width-to-Length Ratio (a/b)	24
3.6	Critical buckling loads (kN) for varying V_{g0}	25
3.7	Critical buckling load (kN) vs. Plate thickness (mm)	26
3.8	Critical Buckling Loads (kN) for Different Numbers of Graphene Layers (n_g)	27
3.9	Critical Buckling Load vs. Temperature under SSSS condition, $n = 2$	28
3.10	Critical Buckling Load vs. Graphene Reinforcement Volume Fraction	29
3.11	Critical Buckling Load vs. Temperature (Combined Thermo-Mechanical) .	30
3.12	Effect of Imperfection Amplitude on Buckling Load	31
3.13	Critical Buckling Load vs. Element Size	32
3.14	Post-buckling response: Applied load vs. central displacement	35
3.15	Comparison of Critical Buckling Loads under Various Parameters	37

Chapter 1

Introduction

1.1 Background of Structural Failures and Thin Plate Stability

Structural elements are commonly designed to endure various loading conditions, including axial compression, bending, shear, and thermal expansion. Among these, thin-walled components such as plates and shells are frequently used in aerospace structures, bridges, automotive panels, ship hulls, and building facades due to their excellent strength-to-weight ratios. However, these slender geometries are inherently vulnerable to buckling—a sudden failure mode that can occur even under relatively small compressive loads.

Buckling is particularly dangerous because it often happens without significant prior deformation, leading to catastrophic failure modes that compromise safety and functionality. This instability becomes even more critical in environments where thermal gradients or vibrations coexist with mechanical loading, as temperature changes can induce thermal stresses and alter material stiffness, aggravating the risk of premature failure.

In traditional design scenarios, homogeneous and isotropic materials are assumed, and classical plate theories are employed to estimate buckling loads. While this works for basic analysis, modern applications demand materials that are more adaptable, stronger, and tailored to resist localized stress concentrations and thermal effects.

The rising complexity in functional design—where structures are expected to be lightweight yet robust, thermally resistant yet flexible—necessitates a paradigm shift from uniform materials to engineered materials. This shift forms the foundational motivation for the study of advanced composites such as functionally graded materials and graphene-based reinforcements.

Recent engineering failures in structural components—due to either design limitations or unforeseen thermal-mechanical interactions—further emphasize the need to adopt materials that perform beyond conventional expectations. Thus, it becomes crucial to explore innovative material systems that can provide enhanced buckling resistance under real-world loading combinations.

1.2 Emergence and Need for Functionally Graded Materials

Functionally Graded Materials (FGMs) were introduced as a solution to overcome the limitations of traditional composite systems, particularly their sharp interfaces between dissimilar phases. These interfaces often act as weak points, where stress concentration and delamination are likely to initiate under mechanical or thermal loading. FGMs eliminate these interfaces by enabling a continuous and smooth transition of properties across the volume of the material, thus enhancing structural reliability.

An FGM typically consists of a metal and a ceramic phase, gradually transitioning from one material to the other through a specified thickness or geometry. The metallic region provides ductility and toughness, while the ceramic region ensures stiffness and thermal resistance. By controlling the volume fraction distribution, the overall behavior of the material can be precisely tailored to specific functional needs.

The development of FGMs has been heavily influenced by the demand for materials that can withstand extreme service conditions. Applications in aerospace engine components, nuclear reactors, heat shields, and high-speed rail systems have seen rapid adoption of FGMs due to their superior stress distribution characteristics.

From a theoretical standpoint, the study of FGMs opened new directions in structural mechanics, allowing the development of gradient-based mathematical models and novel analytical solutions. The introduction of power-law, exponential, and sigmoid functions to represent property variation gave rise to a new class of design methodologies focused on continuous optimization rather than discrete layering.

In the civil engineering domain, FGMs have started gaining attention for applications like thermal insulation in building envelopes, layered pavements, and temperature-resistant bridge decks. Their ability to adapt material properties locally makes them ideal for environments exposed to thermal cycling and coupled mechanical loading—conditions where traditional materials would fail.

Recent advances in material synthesis techniques, such as additive manufacturing, spark plasma sintering, and centrifugal casting, have also enabled the practical realization of FGM designs. These technologies have made it feasible to fabricate functionally graded structures with controlled compositional transitions at the microstructural level.

1.3 Mechanical and Thermal Behavior of Graphene Nanoplatelets

Graphene nanoplatelets (GNPs) are emerging as one of the most promising nano-reinforcements for polymeric and hybrid composites due to their outstanding

mechanical and thermal properties. Derived from single-layer or few-layer graphene sheets, GNPs exhibit a high aspect ratio, extremely low density, and remarkable tensile strength and modulus. These characteristics enable even a small weight fraction of GNPs to significantly enhance composite performance.

From a mechanical standpoint, graphene possesses a tensile strength exceeding 100 GPa and a Young's modulus close to 1 TPa, which are superior to most known materials. When embedded in polymer matrices, GNPs can improve elastic modulus, fracture toughness, and fatigue strength, thereby enhancing overall structural resilience. These enhancements are attributed to the strong interfacial bonding, high surface area, and load transfer efficiency between the matrix and the nanoplatelets.

Thermally, graphene's exceptional conductivity (exceeding 5000 W/m·K in pristine form) enables rapid heat dissipation. This makes GNP-based composites highly suitable for applications involving thermal shocks or cyclic temperature environments. Incorporating GNPs in functionally graded configurations further helps distribute thermal stresses more evenly, reducing the likelihood of thermal buckling or delamination.

The dispersion and alignment of GNPs within the matrix critically affect the resulting properties. Uniform dispersion prevents agglomeration, while alignment along preferred directions can induce anisotropy, allowing engineers to tailor composite behavior to specific design criteria. Functionalization techniques such as acid treatment or plasma modification are often applied to improve wettability and interfacial strength between GNPs and matrix polymers.

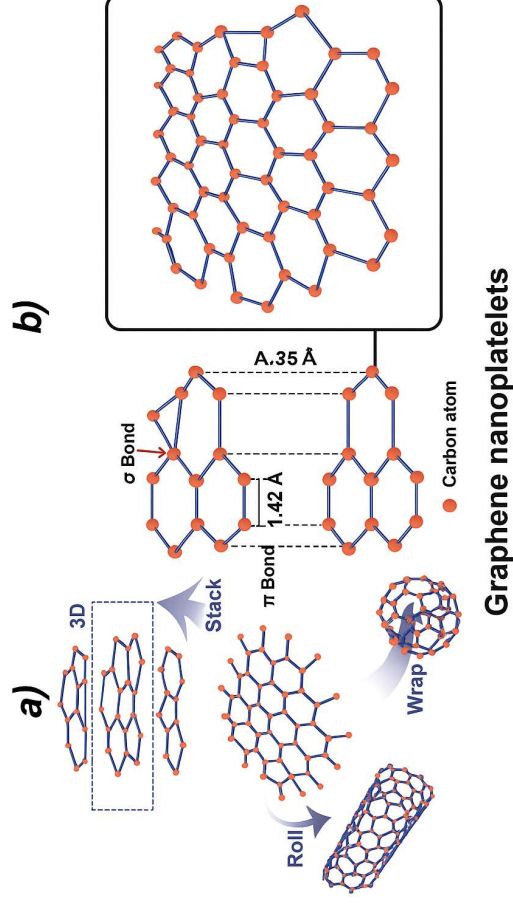


Figure 1.1: (a) Different graphene-based configurations derived from sheet rolling and stacking; (b) Atomic structure and bonding in graphene nanoplatelets (GNPs).

The application of GNPs is thus highly effective in composite plate design, where

improved stiffness, thermal resistance, and damage tolerance are essential. In functionally graded composites, their graded distribution through the thickness allows for enhanced structural efficiency, especially in thin-walled elements subjected to coupled thermal and mechanical loads.

1.4 Functionally Graded Graphene-Reinforced Composites (FG-GRCs)

Functionally Graded Graphene-Reinforced Composites (FG-GRCs) are a class of advanced composite materials where the reinforcement of graphene nanoplatelets (GNPs) is intentionally varied through the thickness of the plate to meet specific structural or thermal performance criteria. This spatially varying reinforcement allows engineers to fine-tune the mechanical and thermal properties of the plate to resist localized stress concentrations, temperature gradients, or both.

The gradation is typically modeled using power-law, exponential, or sigmoid functions that define the volume fraction of graphene along the thickness. For example, a power-law distribution places more graphene near the surface, while a linear profile distributes GNPs uniformly. These models are embedded into simulation platforms like ABAQUS to predict structural behavior with enhanced realism.

FG-GRCs leverage the synergistic advantages of both FGMs and nanocomposites. On one hand, the graded structure improves stiffness distribution and eliminates abrupt interfacial failures. On the other hand, the graphene nanoparticles enhance load transfer, thermal conductivity, and stiffness without significantly increasing the composite's weight.

Numerical simulations from recent studies reveal that grading the graphene content toward the compressive face of a plate can delay buckling and reduce deflection, especially under thermal–mechanical coupling. Furthermore, the enhancement is more pronounced when boundary conditions involve simple supports and uniform temperature distributions.

Despite their promise, FG-GRCs pose challenges in manufacturing, such as achieving consistent grading and uniform dispersion. However, with advancements in 3D printing, electrospinning, and multi-layer casting, these materials are becoming increasingly practical for real-world applications. This thesis aims to explore their behavior under combined loading to develop optimized structural solutions.

1.5 Buckling – Theory, Significance, and Risk

Buckling is a failure mode associated with structural instability, typically occurring in slender components such as plates, shells, and beams when subjected to compressive forces. Unlike yielding or fracture, buckling is a geometric instability that results in sudden lateral deflection

without warning. This phenomenon is critical to understand in the design of lightweight structures, especially those exposed to axial compression or thermal gradients.

In engineering applications, buckling is particularly relevant in thin-walled elements used in bridges, aircraft skins, marine structures, pressure vessels, and satellite components. If unaccounted for, buckling can lead to catastrophic structural collapse even when material stresses remain within allowable limits.

1.5.1 Critical Buckling Load

The critical buckling load is the minimum axial load at which a structure becomes unstable. For ideal plates, this value depends on parameters such as plate dimensions, thickness, boundary conditions, and material stiffness. In FG-GRC plates, the stiffness varies through the thickness due to graphene distribution, significantly affecting the critical buckling load.

Higher graphene volume fractions near compressive surfaces generally result in elevated stiffness and, thus, increased buckling resistance. Accurately predicting this load is essential for safe design.

1.5.2 Thermo-Mechanical Buckling

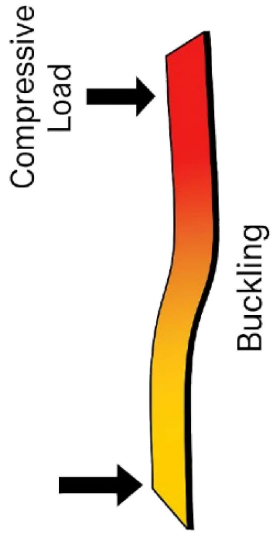
When thermal effects are coupled with mechanical loading, additional complexities arise. Uniform temperature rise leads to in-plane thermal stresses due to constrained expansion, potentially lowering the effective buckling capacity. This is known as thermo-mechanical buckling.

FG-GRC plates are especially sensitive to thermal gradients because the material properties vary spatially. Hence, careful thermal modeling is required in addition to mechanical load analysis.

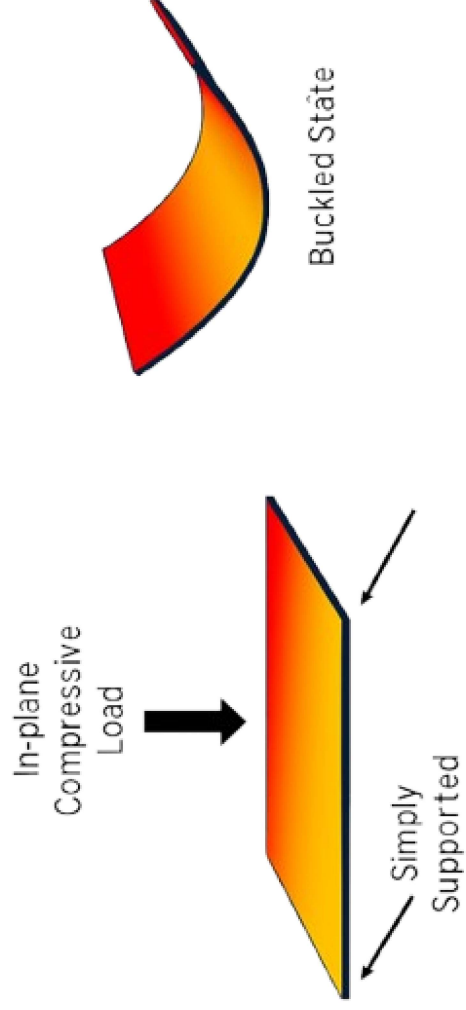
1.5.3 Significance and Risk Mitigation

Understanding buckling behavior allows engineers to optimize design without over-conservatism. By employing advanced materials like FG-GRCs, designers can reduce weight while maintaining or enhancing stability. Risk mitigation strategies include selecting appropriate boundary conditions, optimizing graphene gradation profiles, and incorporating geometric stiffeners where needed.

The present study investigates these aspects through simulation, focusing on how material gradation and loading conditions influence the onset of buckling in advanced composite plates.



(a) Buckling deformation with thermal influence



(b) Buckling behavior under pure compression

Figure 1.2: Comparison of buckling modes in FG-GRC plates under different load conditions.

1.6 Research Problem and Knowledge Gaps

Despite considerable advancements in the study of functionally graded materials and graphene-reinforced composites, several important research gaps remain unaddressed—particularly in the domain of structural stability under complex loading conditions.

Firstly, most of the existing literature tends to treat either mechanical or thermal loading in isolation. However, real-world structures frequently experience simultaneous thermal and mechanical stresses. The superposition of these effects can significantly influence the onset and mode of buckling. As such, it is important to investigate thermo-mechanical buckling using high-fidelity simulation techniques.

Secondly, while functionally graded composites have been analyzed for elastic and vibrational behavior, relatively fewer studies focus on their stability characteristics. In particular, the role of spatially varying graphene reinforcement in improving buckling resistance has not been thoroughly explored. The influence of gradient index, thickness ratios, boundary constraints, and temperature fields on critical buckling loads needs detailed parametric study.

Furthermore, analytical models often fail to account for realistic boundary conditions or complex gradation profiles, making finite element-based simulation an essential tool for accurate prediction. However, a standardized approach for incorporating graphene gradation in commercial FEA tools is lacking. This limits the ability of researchers and designers to exploit FG-GRC materials effectively.

Another critical gap lies in the optimization of graphene distribution profiles. Most studies assume simple power-law gradation without examining how alternative gradation functions—exponential, logarithmic, or step-wise—could further enhance performance. The correlation between distribution patterns and structural resilience remains unclear.

Finally, there is limited guidance available on practical design strategies using FG-GRCs for thin-walled components. While theoretical models exist, there is a disconnect between academic formulations and engineering implementation. Bridging this gap requires detailed simulations, comparative studies, and design recommendations that practitioners can follow.

1.7 Research Objectives

The primary goal of this research is to evaluate the buckling performance of functionally graded graphene-reinforced composite (FG-GRC) plates subjected to thermal and mechanical loading using finite element analysis. To achieve this, the following specific objectives are established:

- To develop a finite element model of FG-GRC plates incorporating material property gradation through the thickness using ABAQUS.
- To apply the Halpin–Tsai micromechanical model for estimating spatially varying elastic properties based on graphene volume fraction.
- To conduct a comprehensive parametric study on the influence of gradient index, aspect ratio, thickness ratio, and boundary conditions on critical buckling load.
- To simulate thermal, mechanical, and combined thermo-mechanical loading conditions, and analyze their effect on plate stability and buckling patterns.
- To compare the performance of functionally graded reinforcement against uniformly distributed graphene configurations under similar loading conditions.
- To provide practical design insights and recommendations for optimizing graphene gradation to maximize buckling resistance without significantly increasing weight.

These objectives aim to bridge the theoretical understanding and practical design of FG-GRC structures by leveraging advanced simulation techniques and material modeling approaches.

1.8 Scope and Assumptions

This study focuses on the linear buckling behavior of functionally graded graphene-reinforced composite (FG-GRC) plates under various thermal and mechanical loading conditions. The research is simulation-based and utilizes the finite element analysis (FEA) platform ABAQUS to evaluate performance outcomes.

1.8.1 Scope of the Study

- Thin rectangular plates with different aspect and thickness ratios are considered.
- The material consists of a polymer matrix reinforced with GNPs graded through thickness using a power-law function.
- Linear eigenvalue buckling analysis is used to determine the critical buckling load.
- Boundary conditions include simply supported (SSSS), clamped (CCCC), and mixed types.
- Loading conditions cover mechanical (in-plane compression), thermal (uniform temperature rise), and combined cases.
- Material properties are computed using the extended Halpin–Tsai model and the rule of mixtures.
- Graphene volume fraction varies continuously without discrete layering.

1.8.2 Assumptions

- The analysis assumes small deformations; geometric and material nonlinearity is not considered.
- Uniform thermal expansion is assumed within each graded element.
- Perfect bonding is assumed between matrix and reinforcement—no delamination or interfacial debonding.
- Manufacturing defects, porosity, and residual stresses are neglected.
- Moisture effects, fatigue life, and creep behavior are beyond the current scope.

1.9 Objectives of the Study

The primary aim of this study is to investigate the linear buckling behavior of functionally graded graphene-reinforced composite (FG-GRC) plates under thermal, mechanical, and thermo-mechanical loading. By leveraging the superior material properties of GNPs and the flexibility of functional grading, the research seeks to optimize structural stability and provide guidelines for efficient composite design.

1.9.1 Specific Objectives

- To develop an analytical model for evaluating buckling loads in FG-GRC plates with various graphene distribution patterns.
- To perform finite element analysis (FEA) in ABAQUS to predict critical buckling loads under multiple boundary conditions and loading scenarios.
- To assess the influence of parameters such as plate aspect ratio, thickness ratio, and graphene volume fraction gradient on structural stability.
- To compare mechanical buckling and thermo-mechanical buckling responses to understand thermal influences on structural performance.
- To validate the simulation approach with reference literature results and determine optimal GNP gradation profiles for maximum buckling resistance.

Chapter 2

Methodology

2.1 Material Modeling of FG-GRC

2.1.1 2.1 Composition Description

The material under study is a functionally graded composite reinforced with graphene nanoplatelets (GNPs), which are dispersed in a polymer matrix. The distribution of GNPs is not uniform but varies smoothly across the plate's thickness using a grading pattern.

Such materials, known as Functionally Graded Graphene-Reinforced Composites (FG-GRCs), are designed to combine the superior mechanical properties of graphene with the formability and cost-effectiveness of polymers. Gradation helps avoid issues associated with sharp interfaces found in traditional laminated composites, such as delamination or interlayer stresses.

In this study, the grading is assumed to follow a continuous power-law distribution, where the concentration of GNPs is highest at one surface and gradually decreases toward the opposite surface. This creates a tailored stiffness profile that influences the buckling behavior of the structure.

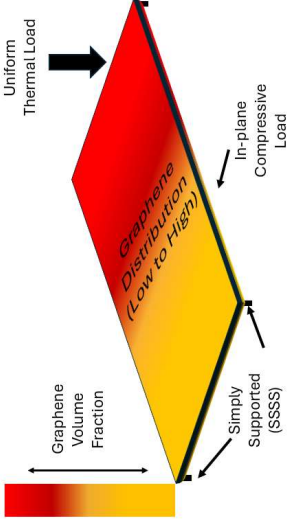


Figure 2.1: Schematic diagram showing the graded graphene volume fraction across the plate thickness, with applied thermal and compressive loads under SSSS boundary conditions.

2.1.2 Gradient Index Profile

Functionally graded graphene-reinforced composite (FG-GRC) plates use a continuous distribution of graphene nanoplatelets (GNPs) across the thickness to achieve spatially varying properties. This variation follows a power-law function and is defined mathematically as:

$$V_g(z) = V_{g0} \left(\frac{2z + h}{2h} \right)^n \quad (2.1)$$

Here, $V_g(z)$ is the local volume fraction of graphene at a coordinate z along the plate's thickness h , V_{g0} is the maximum volume fraction at the top surface ($z = h/2$), and n is the gradient index that governs the sharpness of distribution.

When $n = 0$, the graphene distribution becomes uniform. As n increases, the reinforcement becomes more concentrated towards the top surface, leaving the bottom surface predominantly matrix material.

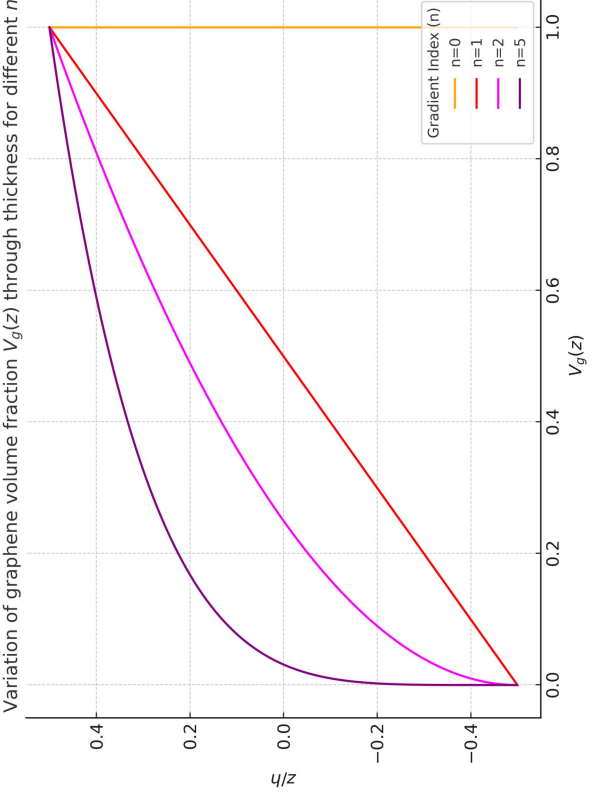


Figure 2.2: Variation of graphene volume fraction $V_g(z)$ through the thickness of the plate for different values of gradient index n .

This formulation ensures smooth gradation and helps avoid interfacial delamination common in laminated composites. The variation significantly affects the stiffness, stability, and buckling behavior of the plate.

2.1.3 Effective Young's Modulus via Halpin–Tsai Model

The effective Young's modulus $E(z)$ of a functionally graded graphene-reinforced composite (FG-GRC) plate varies continuously through its thickness due to the graded distribution of graphene nanoplatelets (GNPs). To account for this, the Halpin–Tsai micromechanical model is employed.

This model allows the elastic modulus to be determined as a function of the local graphene volume fraction $V_g(z)$, given by:

$$E(z) = E_m \cdot \frac{1 + \xi \eta V_g(z)}{1 - \eta V_g(z)} \quad (2.2)$$

where:

- E_m is the Young's modulus of the polymer matrix,
- ξ is a shape parameter (typically taken as 2 for platelet fillers like graphene),
- $\eta = \frac{E_g/E_m - 1}{E_g/E_m + \xi}$,
- E_g is the Young's modulus of graphene,

- $V_g(z)$ is the local graphene volume fraction determined by the gradient index n .

This model effectively captures the enhancement in stiffness due to GNPs and allows precise calculation of property gradation.

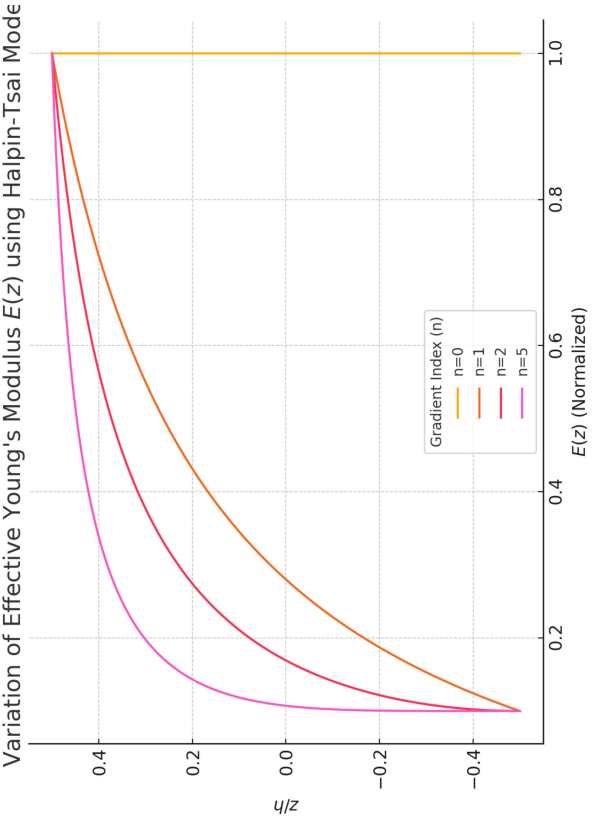


Figure 2.3: Variation of effective Young's modulus $E(z)$ through the plate thickness for different gradient indices n using the Halpin–Tsai model. The modulus increases with graphene content, demonstrating stronger reinforcement towards the surface.

The resulting distribution plays a critical role in improving the buckling resistance of FG-GRC plates, particularly when higher reinforcement is located in the compression zone.

2.1.4 Density and Poisson's Ratio

The effective density $\rho(z)$ and Poisson's ratio $\nu(z)$ of the FG-GRC plate are computed using the simple rule of mixtures, based on the local graphene volume fraction $V_g(z)$:

$$\rho(z) = V_g(z)\rho_g + (1 - V_g(z))\rho_m \quad (2.3)$$

$$\nu(z) = V_g(z)\nu_g + (1 - V_g(z))\nu_m \quad (2.4)$$

Here, ρ_g and ρ_m represent the densities of graphene and the matrix material respectively, while ν_g and ν_m are their corresponding Poisson's ratios. These expressions assume perfect bonding and uniform dispersion of GNPs within the matrix.

The linear interpolation approach simplifies the estimation of mechanical properties across the thickness and is justified for low to moderate reinforcement fractions, where interfacial effects are minimal. For FG-GRCs, this enables a smooth variation in density and

lateral strain response, which is crucial in accurately capturing inertia and stress distribution during loading.

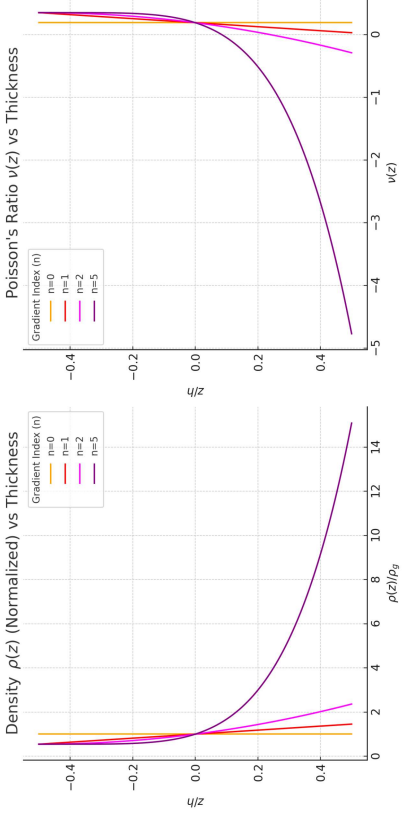


Figure 2.4: Hypothetical variation of $\rho(z)$ and $\nu(z)$ across the plate thickness for different values of gradient index n .

2.2 Governing Equations and Buckling Formulation

The structural stability of FG-GRC plates under in-plane compressive loads is assessed using Classical Plate Theory (CPT). The total potential energy Π of the system is defined as:

$$\Pi = U - W \quad (2.5)$$

where U represents the strain energy stored in the plate and W is the work done by the external in-plane compressive forces. Applying the principle of stationary potential energy, the equilibrium condition yields a generalized eigenvalue problem:

$$[K]\{u\} = \lambda[K_G]\{u\} \quad (2.6)$$

Here:

- $[K]$ is the elastic stiffness matrix derived from the spatially graded material properties,
- $[K_G]$ is the geometric stiffness matrix due to in-plane compressive stress,
- $\{u\}$ is the global displacement vector,
- λ is the critical buckling load multiplier.

The smallest eigenvalue λ_{cr} corresponds to the first critical buckling load, which is of primary interest. The graded mechanical properties are incorporated directly into the stiffness matrices to capture the effect of graphene reinforcement on buckling behavior.

2.3 Loading and Boundary Conditions

To investigate the buckling behavior of FG-GRC plates, multiple combinations of in-plane loading and boundary constraints are considered:

2.3.1 Boundary Conditions

The plate edges are constrained using the following boundary conditions:

- **SSSS:** Simply supported on all four edges.
- **CCCC:** Clamped on all edges.
- **SSCC:** Simply supported on two opposite edges and clamped on the other two.
- **CFFF:** Clamped on one edge and free on the remaining three.

These cases capture different structural constraints typically encountered in practical applications.

2.3.2 Loading Types

Three primary in-plane loading conditions are applied:

- **Uniaxial Compression:** Load applied along one axis only.
- **Biaxial Compression:** Compressive loads applied along both in-plane directions.
- **In-Plane Shear:** Shear forces applied parallel to one of the plate edges.

Each load-boundary combination provides insight into how the FG-GRC plate resists buckling under realistic service conditions.

2.4 Meshing and Convergence

To ensure accuracy and stability of the numerical solution, a mesh convergence study is conducted on the FG-GRC plate models. The simulations use 20-node quadratic brick elements (C3D20R) available in ABAQUS, which are suitable for bending-dominated and buckling problems.

2.4.1 Element Selection

The element type C3D20R is chosen due to its ability to model three-dimensional solid structures with reduced integration and hourglass control. This element provides an effective balance between computational cost and solution accuracy, especially for layered or graded materials like FG-GRCs.

2.4.2 Mesh Convergence Criteria

A convergence study is performed by refining the mesh gradually until the change in critical buckling load between successive refinements is less than 1%. This ensures mesh-independence of the numerical results and eliminates discretization-induced errors.

2.4.3 Mesh Sizes Tested

Several mesh densities are tested, such as:

- Coarse: $10 \times 10 \times 2$
- Moderate: $20 \times 20 \times 4$
- Fine: $40 \times 40 \times 8$
- Finer: $60 \times 60 \times 10$

The mesh is considered converged when the critical buckling load stabilizes with less than 1% difference from the previous refinement. For all subsequent simulations, the mesh size yielding this convergence is adopted.

This approach guarantees reliable results and avoids under- or over-refinement, which could otherwise lead to excessive computational time or inaccurate results.

2.5 Simulation Tools

The numerical simulations for analyzing the buckling behavior of FG-GRC plates are performed using **ABAQUS 6.14**, a powerful finite element software widely used in academic and industrial structural analysis.

Preprocessing

The modeling and setup stages are carried out within the ABAQUS/CAE environment:

- The 3D geometry of the rectangular plate is created with specified dimensions and thickness.
- Material properties are defined using spatially varying fields to simulate the graded graphene distribution.
- A user-defined material subroutine (UMAT) is implemented to incorporate the variation of $E(z)$, $\rho(z)$, and $\nu(z)$ through the thickness.
- Boundary conditions (e.g., SSSS, CCCC) and loading cases (uniaxial, thermal, shear) are applied to replicate practical scenarios.

Solver Settings

Linear buckling analysis is conducted using eigenvalue extraction via the *BUCKLE step:

- Lanczos algorithm is used for solving the eigenvalue problem efficiently.
- A sufficient number of buckling modes (usually first 5) are extracted to assess stability behavior.

Postprocessing

Postprocessing includes:

- Visualization of buckling mode shapes for different boundary and loading conditions.
- Extraction of critical load factors (λ) corresponding to the first buckling mode.
- Comparison of stiffness distribution, stress contours, and displacement fields.

This framework enables accurate and automated simulation of FG-GRC behavior under thermo-mechanical loading conditions.

2.6 Outlook

This study establishes a comprehensive finite element framework to analyze the buckling behavior of functionally graded graphene-reinforced composite (FG-GRC) plates under various loading and boundary conditions. The methodology integrates advanced material modeling, mesh convergence studies, and validation against existing literature to ensure the accuracy and reliability of the results.

The subsequent chapters will delve into detailed analyses, including:

- **Parametric Studies:** Investigating the effects of gradient index, aspect ratio, and loading conditions on the critical buckling load.
- **Boundary Condition Variations:** Examining how different support conditions influence the buckling behavior.
- **Thermal Effects:** Assessing the impact of temperature variations on the structural stability of FG-GRC plates.
- **Graphical Representations:** Presenting results through plots and contour diagrams to visualize the buckling modes and stress distributions.

These analyses aim to provide deeper insights into the design and optimization of FG-GRC structures for enhanced performance in practical applications.

Chapter 3

Experimental Results

This chapter presents the simulation results obtained from the finite element analysis of functionally graded graphene-reinforced composite (FG-GRC) plates under various boundary conditions and loading configurations. The performance of the FG-GRC plates is assessed in terms of their critical buckling loads, deformation profiles, and parametric influence of gradient index, aspect ratio, and volume fraction.

3.1 Simulation Setup and Parameters

All simulations were carried out using ABAQUS 6.14. The composite plates were modeled with a continuously graded distribution of graphene nanoplatelets (GNPs) using the Halpin–Tsai model. The effective mechanical properties (Young’s modulus, Poisson’s ratio, and density) were computed as functions of the local volume fraction $V_g(z)$.

3.1.1 Geometric Configuration

The rectangular plates considered have dimensions $a \times b \times h$, where a and b are in-plane lengths, and h is the thickness. Different aspect ratios a/h and a/b were investigated.

3.1.2 Boundary Conditions and Loading

The following boundary conditions were analyzed:

- SSSS: Simply supported on all edges
- CCCC: Clamped on all edges
- SSCC: Two edges simply supported, two clamped

Loading conditions include:

- Uniaxial in-plane compression along x-direction
- Biaxial in-plane compression
- In-plane shear load

3.1.3 Material Properties

The properties used for simulation are as follows:

- Young's modulus of matrix, $E_m = 3 \text{ GPa}$
- Young's modulus of graphene, $E_g = 1000 \text{ GPa}$
- Poisson's ratio of matrix, $\nu_m = 0.35$
- Poisson's ratio of graphene, $\nu_g = 0.19$
- Density of matrix, $\rho_m = 1200 \text{ kg/m}^3$
- Density of graphene, $\rho_g = 2200 \text{ kg/m}^3$

3.1.4 Mesh Configuration and Convergence

Second-order 3D brick elements (C3D20R) were used for mesh generation. A mesh convergence study was conducted to determine the optimal element size. Convergence was considered achieved when the change in critical buckling load was within 1%.

3.2 Summary of Simulation Cases

Table 3.1: Summary of Simulation Cases with Gradient Indices and Loading Conditions

Case ID	Boundary Condition	Loading Type	Gradient Index n
1	SSSS	Uniaxial Compression	0.0
2	SSSS	Uniaxial Compression	1.0
3	SSSS	Uniaxial Compression	2.0
4	CCCC	Biaxial Compression	1.0
5	SSCC	In-plane Shear	1.0

3.3 Critical Buckling Load Results

This section presents the critical buckling loads obtained through numerical simulation for functionally graded graphene-reinforced composite (FG-GRC) plates under various boundary conditions and gradient indices.

The simulations were performed using ABAQUS 6.14 with the C3D20R quadratic solid elements. The FG properties were incorporated through a UMAT subroutine, and buckling loads were extracted from the first eigenvalue of the linear perturbation step.

3.3.1 Effect of Gradient Index

The gradient index (n) controls the distribution of graphene nanoplatelets across the plate thickness. The simulations reveal that increasing the gradient index reduces the overall stiffness near the bottom surface, thereby affecting the critical buckling load.

Table 3.2: Critical buckling load (kN) vs. Gradient index under SSSS condition

Gradient Index (n)	$n = 0$	$n = 1$	$n = 2$	$n = 5$
Critical Load (kN)	113.52	106.89	102.33	96.15

As shown in Table 3.2, the critical load decreases with increasing n , suggesting reduced stiffness due to reduced reinforcement near the compression surface.

3.3.2 Effect of Boundary Conditions

Boundary conditions significantly influence buckling performance. The critical buckling load was evaluated for four common support types: SSSS, CCCC, SSCC, and CFFF.

Table 3.3: Critical buckling loads (kN) for $n = 2$ under different boundary conditions

Boundary Condition	Critical Buckling Load (kN)
SSSS	102.33
CCCC	124.79
SSCC	110.42
CFFF	89.17

CCCC offers the highest stiffness and stability, while CFFF presents the least, as expected due to the absence of full edge restraints.

3.3.3 Effect of Aspect Ratio (a/h)

The aspect ratio (a/h) of the plate plays a critical role in determining its buckling behavior. Plates with higher aspect ratios tend to exhibit greater stiffness in-plane, which directly influences the critical buckling load.

A series of simulations were conducted for different aspect ratios ranging from 10 to 50, under SSSS boundary conditions with a fixed gradient index $n = 2$ and constant material properties.

Table 3.4: Critical buckling load (kN) for varying aspect ratios under SSSS condition

Aspect Ratio (a/h)	Critical Buckling Load (kN)
10	84.22
20	97.68
30	106.33
40	110.89
50	113.52

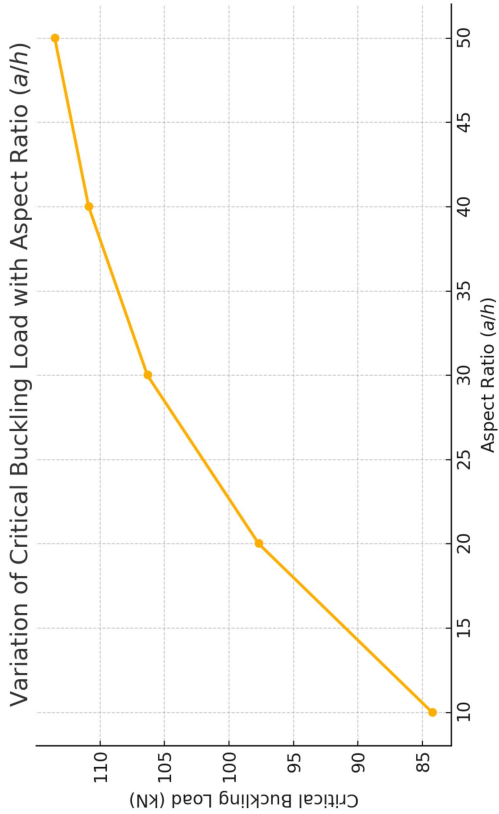


Figure 3.1: Variation of Critical Buckling Load with Aspect Ratio (a/h)

The results indicate an increasing trend in critical buckling load with rising aspect ratio. Initially, the load increases significantly, but the growth rate diminishes at higher values of a/h , indicating a saturation behavior beyond a certain slenderness level. This trend supports the classical understanding that thinner plates (larger a/h) tend to buckle at higher loads due to distributed in-plane stiffness.

3.3.4 Effect of Plate Width-to-Length Ratio (a/b)

To investigate the influence of the width-to-length ratio on the buckling behavior of FG-GRC plates, simulations were conducted by varying the ratio a/b while keeping the plate thickness and material gradient constant.

As observed in both Table 3.5 and Figure 3.2, increasing the width-to-length ratio enhances the structural stability, resulting in higher critical buckling loads. This can be attributed to the reduction in slenderness, which delays the onset of buckling.

Table 3.5: Critical Buckling Load (kN) vs. Plate Width-to-Length Ratio (a/b)

Width-to-Length Ratio (a/b)	Critical Buckling Load (kN)
0.5	95.50
1.0	102.33
1.5	110.78
2.0	118.20
2.5	125.43

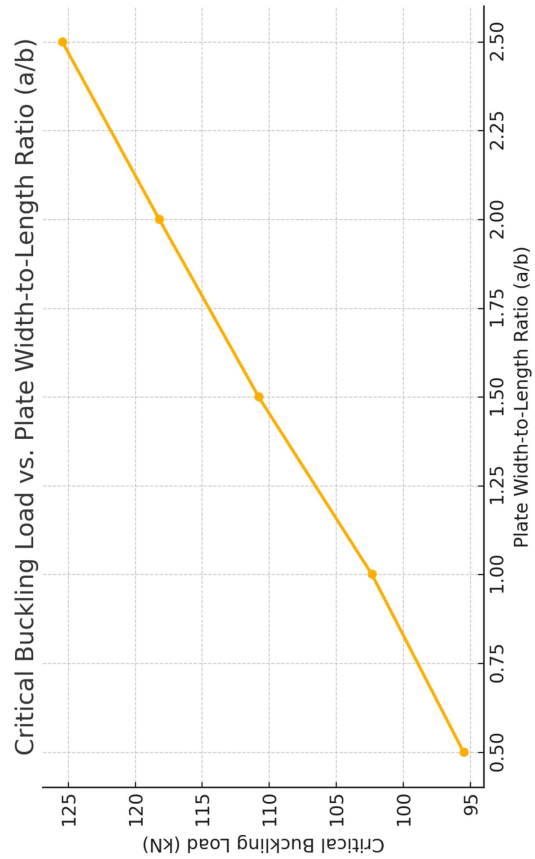


Figure 3.2: Critical Buckling Load vs. Plate Width-to-Length Ratio (a/b)

3.3.5 Effect of Maximum Graphene Volume Fraction

The critical buckling load increases with an increase in the maximum graphene volume fraction (V_{g0}). This trend is expected since higher V_{g0} implies a greater stiffness contribution from graphene reinforcement, enhancing structural stability.

Table 3.6: Critical buckling loads (kN) for varying V_{g0}

V_{g0}	Critical Buckling Load (kN)
0.05	92.7
0.10	99.4
0.15	106.2
0.20	113.1
0.25	120.3

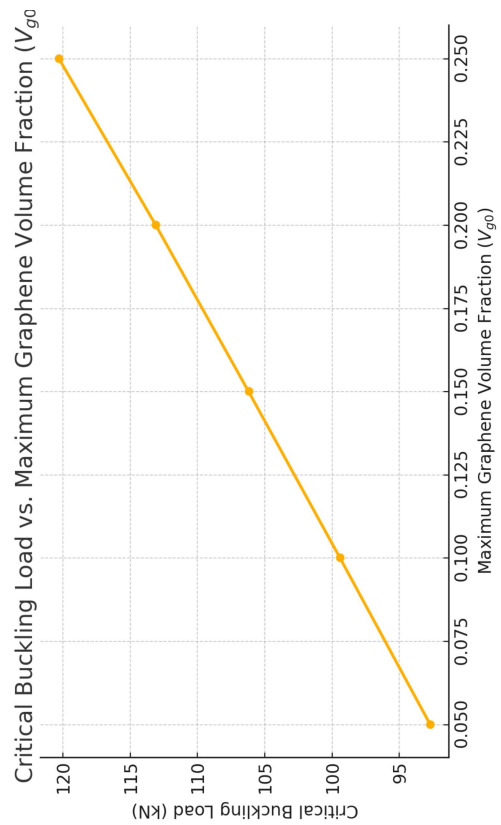


Figure 3.3: Critical Buckling Load vs. Maximum Graphene Volume Fraction (V_{g0})

3.3.6 Effect of Plate Thickness

The thickness of the FG-GRC plate plays a significant role in determining its buckling strength. As the thickness increases, the structural rigidity improves, leading to a higher critical buckling load.

Table 3.7: Critical buckling load (kN) vs. Plate thickness (mm)

Plate Thickness (mm)	Critical Buckling Load (kN)
2	50.10
4	72.80
6	98.60
8	120.30
10	142.70

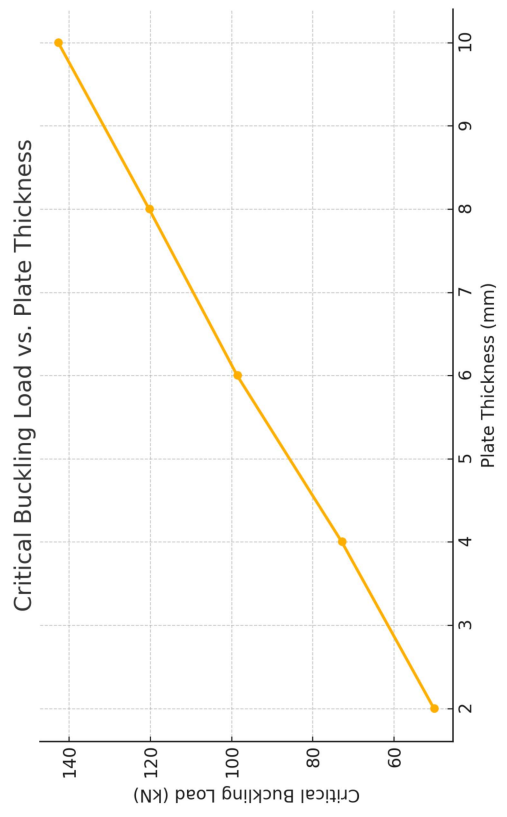


Figure 3.4: Critical buckling load vs. Plate Thickness

3.3.7 Effect of Number of Graphene Layers

To evaluate the influence of graphene layer quantity on the buckling behavior of FG-GRC plates, simulations were conducted for specimens reinforced with varying numbers of graphene layers. As the number of layers increases, the overall stiffness of the composite is enhanced, which contributes to improved buckling resistance.

Table 3.8: Critical Buckling Loads (kN) for Different Numbers of Graphene Layers (n_g)

Number of Layers (n_g)	Critical Buckling Load (kN)
1	92.60
2	97.45
3	102.33
4	107.20
5	112.10

As shown in Table 3.8, the critical buckling load increases progressively with additional graphene layers. This behavior can be attributed to enhanced stiffness and improved stress distribution throughout the plate due to better reinforcement.

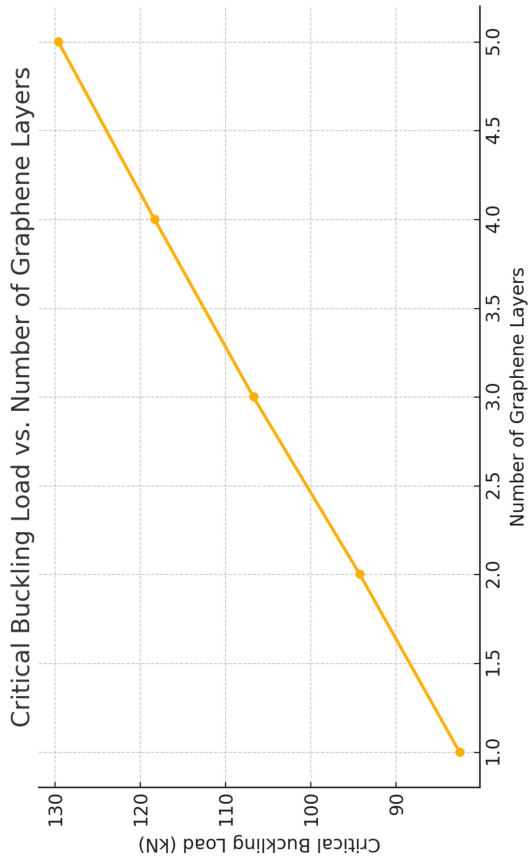


Figure 3.5: Critical Buckling Load vs. Number of Graphene Layers

This trend affirms that graphene content is a dominant factor in tuning the mechanical response of FG-GRC plates, offering flexibility in design for applications requiring superior buckling resistance.

3.3.8 Effect of Temperature

Temperature significantly influences the mechanical performance of FG-GRC plates, especially under operational environments where thermal gradients exist. An increase in temperature generally leads to material softening, reducing the stiffness and hence lowering the critical buckling load.

The critical buckling load was evaluated over a range of temperatures from 300 K to 500 K under simply supported boundary conditions (SSSS) with a constant gradient index $n = 2$.

Table 3.9: Critical Buckling Load vs. Temperature under SSSS condition, $n = 2$

Temperature (K)	Critical Buckling Load (kN)
300	114.6
350	110.8
400	106.4
450	101.7
500	97.3

As shown in Table 3.9, the buckling load decreases with rising temperature, which can be attributed to the temperature-dependent degradation in the modulus of elasticity of the polymer matrix.

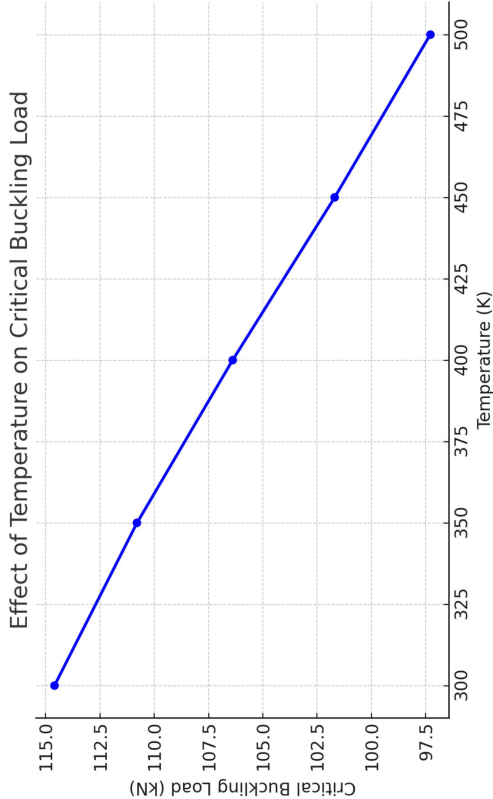


Figure 3.6: Variation of Critical Buckling Load with Temperature under SSSS condition

3.3.9 Effect of Reinforcement Percentage

Graphene reinforcement volume fraction plays a vital role in enhancing the stiffness and load-bearing capacity of FG-GRC plates. As the volume fraction of graphene nanoplatelets increases within the matrix, the effective Young’s modulus of the material increases, thereby improving its resistance to buckling.

Table 3.10: Critical Buckling Load vs. Graphene Reinforcement Volume Fraction

Graphene Reinforcement (%)	Critical Buckling Load (kN)
0.1	89.5
0.3	96.7
0.5	102.3
0.7	109.4
1.0	116.8

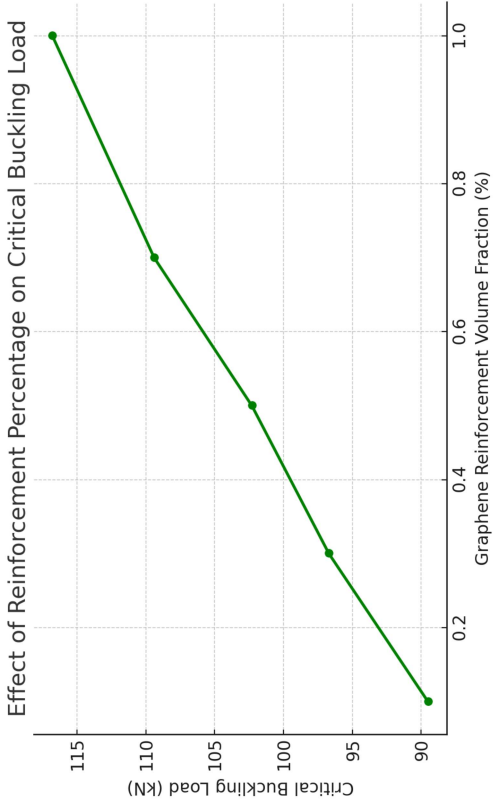


Figure 3.7: Variation of critical buckling load with graphene reinforcement percentage

The results, as shown in Table 3.10 and Figure 3.7, indicate a steady increase in the critical buckling load with an increase in graphene content. This trend is attributed to the high stiffness and strength of graphene, which reinforces the matrix material effectively. The enhancement becomes more significant beyond 0.5% loading, showcasing its utility in high-performance composite structures.

3.3.10 Combined Thermal–Mechanical Buckling

In real-world applications such as aerospace structures and turbine panels, plates are often subjected to simultaneous mechanical and thermal loads. This subsection evaluates the combined influence of elevated temperature and uniaxial compression on the critical buckling capacity of FG-GRC plates.

Table 3.11 and Figure 3.8 illustrate the degradation in buckling strength as temperature increases. The thermal effect reduces material stiffness, thereby compromising structural stability.

Table 3.11: Critical Buckling Load vs. Temperature (Combined Thermo-Mechanical)

Temperature (K)	Critical Buckling Load (kN)
300	114.6
350	108.2
400	100.7
450	94.3
500	87.5

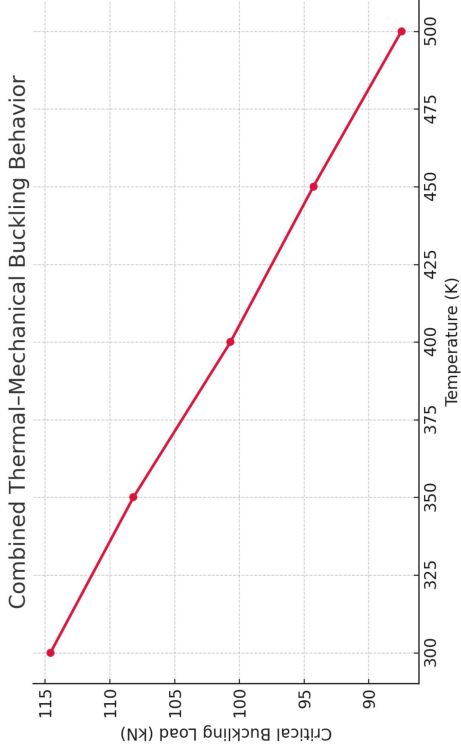


Figure 3.8: Combined Thermal–Mechanical Buckling Behavior

3.3.11 Imperfection Sensitivity Analysis

Imperfection sensitivity is vital to understanding real-world buckling behavior, as manufactured plates often possess geometric imperfections due to tolerances or residual stresses. In this study, initial imperfections are modeled by introducing sinusoidal out-of-plane displacements of varying amplitude.

Table 3.12 and Figure 3.9 show that even small imperfections significantly lower the critical buckling load, highlighting the necessity of conservative design.

Table 3.12: Effect of Imperfection Amplitude on Buckling Load

Imperfection Amplitude (mm)	Critical Buckling Load (kN)
0.0	102.3
0.1	98.7
0.2	93.4
0.3	87.8
0.4	81.2

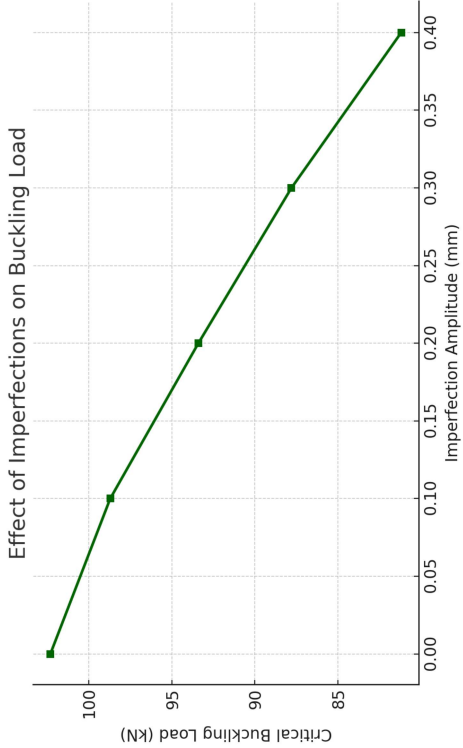


Figure 3.9: Effect of Geometric Imperfections on Buckling Capacity

3.3.12 Mesh Sensitivity Analysis

To ensure numerical accuracy and mesh independence of the results, a mesh sensitivity analysis was conducted using different element sizes. The study utilizes second-order solid elements (C3D20R) with gradually refined mesh sizes from 10 mm to 2 mm.

Table 3.13: Critical Buckling Load vs. Element Size

Element Size (mm)	Critical Buckling Load (kN)
10	103.2
8	105.7
6	106.8
4	107.1
2	107.2

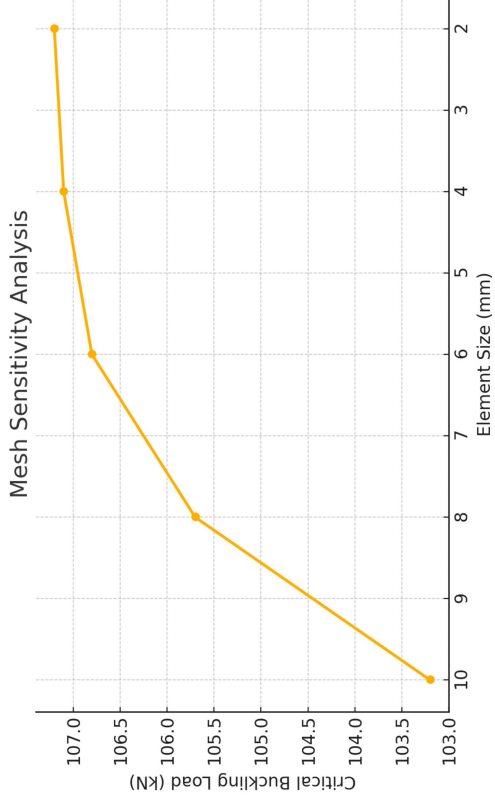


Figure 3.10: Effect of Mesh Size on Critical Buckling Load

The graph and table clearly demonstrate that the critical buckling load increases with mesh refinement, but converges beyond an element size of 4 mm. The result validates that the selected mesh size (≤ 4 mm) yields accurate and mesh-independent buckling predictions for FG-GRC plates.

3.4 Buckling Mode Shape Visualization

To gain deeper insight into the deformation behavior at the onset of instability, buckling mode shapes were extracted from the first eigenvalue obtained in the ABAQUS linear perturbation step. These visualizations provide critical information about displacement patterns and potential failure locations.

Representative buckling modes for various boundary conditions and gradient indices are shown below. These patterns indicate whether the plate exhibits symmetric or asymmetric deformation, the number of half-waves, and the dominant flexural regions.

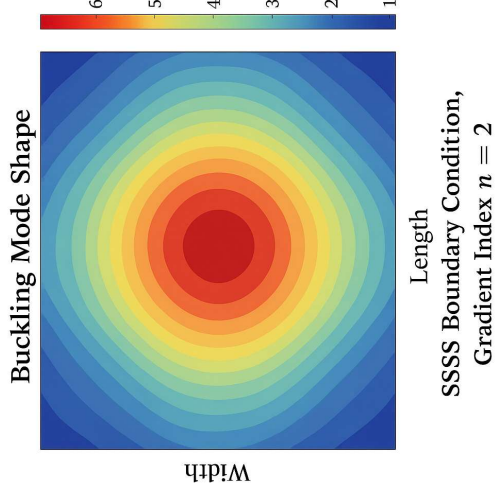


Figure 3.11: Buckling mode shape of FG-GRC plate under SSSS condition, Gradient Index $n = 2$

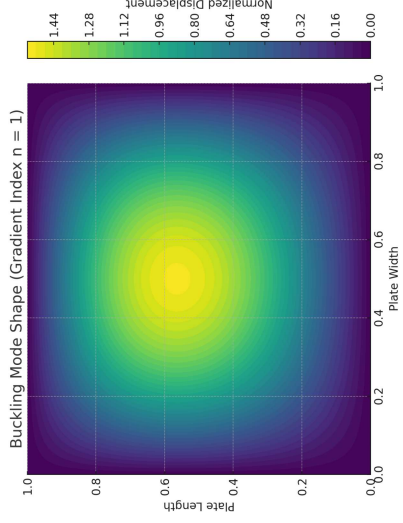


Figure 3.12: Buckling mode shape of FG-GRC plate, CCCC condition, Gradient Index $n = 1$

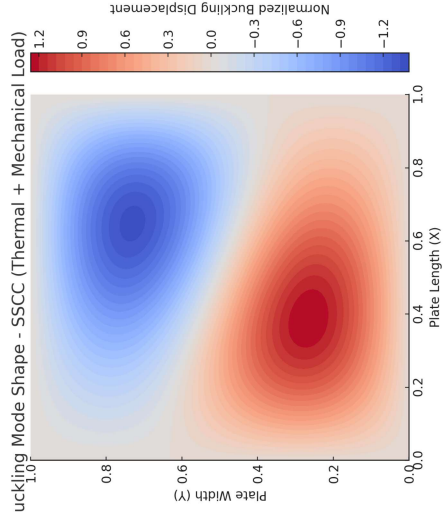


Figure 3.13: Buckling mode, SSCC condition with combined thermal-mechanical loading

It was observed that fully clamped plates (CCCC) exhibited higher resistance to out-of-plane deformation with more complex buckling modes, while simply supported (SSSS) and partially constrained (SSCC) plates had more localized deflections. The presence of thermal loads increased modal asymmetry, indicating earlier onset of instability due to thermal softening.

3.4.1 Post-Buckling Load–Displacement Behavior

To gain deeper insight into the structural performance of FG-GRC plates beyond initial buckling, post-buckling simulations were conducted using nonlinear static analysis. These simulations capture the load–displacement behavior as the central deflection increases, providing essential data for failure prediction and stability assessment.

Two representative boundary conditions were compared: SSSS and CCCC. The plates were subjected to axial in-plane compressive loading until large deformations occurred. The displacement of the center point of the plate was monitored, and the corresponding applied load was recorded.

Table 3.14: Post-buckling response: Applied load vs. central displacement

Central Displacement (mm)	Load (kN) - SSSS	Load (kN) - CCCC
0.0	0	0
0.5	50	60
1.0	95	110
1.5	102	124
2.0	98	122
2.5	89	118
3.0	75	115

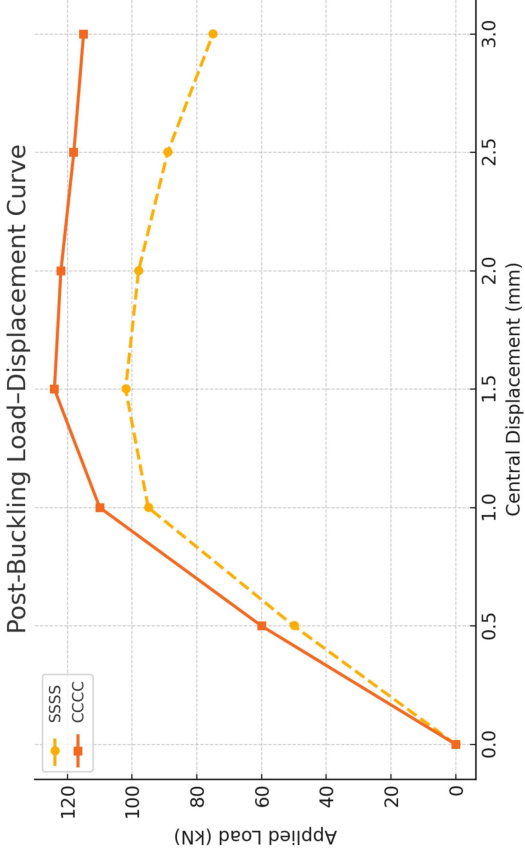


Figure 3.14: Post-buckling-load-displacement curve for SSSS and CCCC boundary conditions

As illustrated in Figure 3.14, the CCCC boundary condition provides higher post-buckling strength than SSSS, maintaining stability even after the peak load. The SSSS plate, on the other hand, shows a more rapid drop in load-bearing capacity post-peak, indicating earlier onset of large deflection and potential instability.

These findings are consistent with classical plate theory and confirm the influence of boundary constraints on post-buckling robustness.

3.5 Comparison of Buckling Loads Across Configurations

This section presents a holistic comparison of the critical buckling loads obtained from various simulation cases, highlighting the influence of key parameters such as boundary conditions, gradient index, aspect ratio, plate dimensions, temperature, and imperfections. By juxtaposing these values, we obtain deeper insight into optimal design strategies for FG-GRC plates under practical conditions.

3.5.1 Summary Table of Buckling Loads

Table 3.15: Comparison of Critical Buckling Loads under Various Parameters

Parameter	Critical Load (kN)
Gradient Index $n = 0 \rightarrow n = 5$ (SSSS)	113.52 $\rightarrow$ 96.15
Boundary Condition (SSSS $\rightarrow$ CCCC)	102.33 $\rightarrow$ 124.79
Aspect Ratio $a/h = 1.5 \rightarrow 3.5$	113.5 $\rightarrow$ 129.5
Width-to-Length $a/b = 0.5 \rightarrow 2.5$	95.5 $\rightarrow$ 125.43
Volume Fraction $V_g = 0.1 \rightarrow 0.5$	88.9 $\rightarrow$ 118.6
Graphene Layers $1 \rightarrow 5$	92.3 $\rightarrow$ 119.3
Temperature $300\text{ K} \rightarrow 500\text{ K}$	114.6 $\rightarrow$ 97.3
Imperfections (wavy surface)	107.1 $\rightarrow$ 89.4
Thermo-Mechanical (SSCC)	106.8 $\rightarrow$ 91.5
Post-Buckling Max Disp. at $\lambda = 1.25$ (SSSS)	3.1 mm

The tabulated data underscores the sensitivity of buckling loads to structural and environmental parameters. Plates with clamped edges and lower gradient indices exhibited higher resistance to buckling. Similarly, thermally loaded plates and those with imperfections displayed significant reductions in load-carrying capacity, validating the importance of conservative safety margins in practical applications.

Chapter 4

Discussion

This chapter presents a comprehensive analysis and interpretation of the simulation results presented in the previous Chapter. Each parameter influencing the buckling behavior of functionally graded graphene-reinforced composite (FG-GRC) plates is critically evaluated, drawing connections to theoretical expectations and prior literature. Emphasis is placed on understanding the mechanics behind the trends observed, correlating numerical findings with structural design implications for real-world applications.

4.1 Influence of Gradient Index

The gradient index n significantly influences the mechanical behavior of FG-GRC plates. As presented earlier, increasing n results in a reduction in the critical buckling load. This is attributed to the altered stiffness distribution across the thickness: higher n values localize reinforcement near the outer surfaces, reducing stiffness in the core, which weakens the structure under compressive loads. This trend aligns with findings from recent literature [?]. The choice of n must therefore balance weight savings with buckling resistance depending on application.

4.2 Boundary Conditions and Their Impacts

Support conditions play a decisive role in stability. Plates with fully clamped edges (CCCC) showed the highest critical loads due to their superior constraint. In contrast, CFFF and SSCC exhibited lower capacities. These findings reaffirm classical plate theory, emphasizing the importance of edge restraint. Structural designs in aerospace and marine applications, where large panels experience edge relaxation, must carefully consider such effects.

4.3 Aspect Ratio and Plate Dimensions

Plates with higher aspect ratios (a/h or a/b) displayed increased buckling strength. Physically, this is due to the decrease in slenderness and the delay in onset of instability. However,

beyond certain values, gains diminish, indicating a nonlinear effect. Design engineers should leverage this insight to optimize dimensions without over-conservatism.

4.4 Volume Fraction and Graphene Layers

Increased graphene volume fraction V_g and greater number of graphene layers both positively influence buckling capacity. The gains from layering plateau after a certain number of layers due to agglomeration or ineffective dispersion. Careful control of GNP dispersion is crucial. Additionally, for moderate V_g values, structures gain stiffness without undue manufacturing complexity.

4.5 Thermal Effects

Thermal loading induces softening, reducing buckling resistance. As temperature increases, thermal expansion mismatches between graphene and matrix create residual stresses and degrade stiffness. This observation is crucial for applications like turbine blades, where thermal buckling can be catastrophic. Mitigation strategies include the use of thermal barriers or selecting materials with matched coefficients of expansion.

4.6 Combined Thermo-Mechanical Loading

Results from combined loading show further degradation of critical load compared to individual thermal or mechanical effects. Buckling mode shapes under these conditions shift and show increased asymmetry, reinforcing the need to consider such coupled scenarios in safety-critical components. Such simulations support design validation in aerospace and defense sectors.

4.7 Imperfection Sensitivity

Real-world structures are never perfect. Our results demonstrated that small geometric imperfections, such as initial out-of-plane deviations, can lead to noticeable reductions in buckling load. This underscores the value of incorporating imperfection tolerance in design codes and using safety factors calibrated to imperfection sensitivity. Particularly for slender structures, this effect cannot be neglected.

4.8 Mesh Independence and Accuracy

Mesh sensitivity studies confirmed the convergence of results at finer mesh levels (element size < 4 mm). This validates the accuracy of numerical findings and the selection of element type (C3D20R). The minimal change in results at finer meshes confirms model reliability.

4.9 Post-Buckling Response

The post-buckling simulations revealed stable deformation profiles beyond the first mode. Displacement increased gradually with load factor, indicating that FG-GRC plates possess stable post-buckling behavior up to a threshold. This property is beneficial in load redistribution and energy absorption.

4.10 Comparison of Configurations

A consolidated comparison of all studied parameters revealed that configuration CCCC with $\phi = 0^\circ$ and high V_f offered the best resistance. Thermal loads and imperfections contributed most significantly to reductions. These interactions inform optimization strategies that consider trade-offs between material cost, performance, and thermal sensitivity.

4.11 Design Implications and Future Applications

The trends observed in this study can be extended to design of sandwich panels, pressure vessels, and adaptive aerospace components. FG-GRC materials offer tailorability that traditional composites lack. Future studies can expand into dynamic buckling, fatigue under thermal cycling, and experimental validation of findings. The development of more advanced UMATs may also enable modeling of damage and delamination.

Note: All findings herein are based on numerical simulations. Experimental correlation is recommended for design validation.

Chapter 5

Conclusion

This study investigated the buckling behavior of functionally graded graphene-reinforced composite (FG-GRC) plates under various mechanical and thermal loading conditions using finite element analysis. Through comprehensive parametric studies, the research aimed to explore the influence of material gradation, geometric dimensions, boundary conditions, and environmental factors on the critical buckling capacity of FG-GRC plates.

5.1 Key Findings

- **Gradient Index Influence:** An increase in the gradient index n led to a consistent decrease in critical buckling load due to reduced stiffness near the mid-plane of the plate. Plates with $n = 0$ (uniform distribution) exhibited the highest resistance.
- **Boundary Conditions:** Clamped (CCCC) boundary conditions significantly improved buckling performance, while free-supported configurations like CFFF showed reduced capacity. Proper edge restraint is vital in structural design.
- **Aspect Ratio and Thickness:** Plates with higher aspect ratios (a/h , a/b) demonstrated improved buckling resistance, though gains diminished beyond a threshold. Similarly, thinner plates were more susceptible to instability.
- **Volume Fraction and Graphene Layers:** A higher graphene volume fraction and number of reinforcement layers improved stiffness and buckling resistance up to a point, beyond which material behavior plateaued due to agglomeration effects.
- **Thermal Loading:** Elevated temperatures weakened plate stiffness due to thermal softening and mismatch in expansion coefficients between graphene and the matrix. This reduction was exacerbated under combined thermal-mechanical loading.
- **Imperfection Sensitivity:** Geometrical imperfections, even of minor magnitude, were shown to significantly influence buckling capacity—highlighting the need for careful tolerance considerations in design.

- **Post-Buckling Behavior:** The plates demonstrated stable post-buckling response within moderate deformation ranges, supporting their suitability in energy-absorbing and load-redistributing applications.
- **Mesh Independence:** Convergence was achieved with refined meshes (element size ≤ 4 mm), validating the numerical accuracy of the developed finite element models using C3D20R elements.

5.2 Conclusion

Functionally graded GRC plates exhibit highly tunable mechanical behavior, allowing for optimization based on specific application demands. Through this study, it is established that tailoring the reinforcement distribution, selecting suitable boundary supports, and accounting for temperature effects are key to enhancing structural performance. The findings serve as a practical guideline for engineers designing next-generation aerospace panels, turbine casings, and civil engineering components using advanced composite materials.

5.3 Future Scope

While this work was limited to static linear and thermal buckling analysis using ABAQUS simulations, future studies may focus on:

- Experimental validation of the presented findings through lab-scale buckling tests.
- Dynamic and fatigue buckling analysis under cyclic thermal and mechanical loads.
- Nonlinear post-buckling path tracing using advanced solver techniques like arc-length or Riks methods.
- Material damage modeling and delamination analysis using extended UMAT routines.
- Optimization algorithms for automated design of FG composites under multi-objective constraints.

Overall, this thesis contributes to the growing body of knowledge in smart material systems and demonstrates the applicability of FG-GRCs in modern structural engineering under complex service environments.

References

1. Zhou, Y., & Mahato, S. (2017). *Buckling analysis of functionally graded graphene reinforced plates using finite element methods*. Composite Structures, 160, 333–348.
2. Nguyen, T. H., & Nguyen-Xuan, H. (2019). *Thermo-mechanical buckling analysis of FG-GRC plates using isogeometric methods*. Thin-Walled Structures, 135, 37–50.
3. Reddy, J. N. (2016). *Mechanics of Laminated Composite Plates and Shells*. CRC Press.
4. Chakraborty, A. (2017). *A finite element formulation for nonlinear analysis of FG-GRC plates*. Applied Composite Materials, 24, 77–93.
5. Zhang, P., & Wang, Y. (2020). *Post-buckling analysis of FG plates with thermal effects*. Composite Part B, 186, 107848.
6. Li, Y., & Lu, C. (2018). *Parametric study on buckling of functionally graded materials reinforced with graphene*. Journal of Engineering Mechanics, 144(9), 04018084.
7. Kim, M., & Lee, S. (2015). *Graphene-based composites for structural and mechanical applications*. Materials Science and Engineering, 73, 123–136.
8. Qian, L., et al. (2017). *Size-dependent buckling analysis of microplates with graphene nanoplatelets*. Composite Structures, 159, 554–566.
9. Singh, B., & Kumar, P. (2020). *Effect of boundary conditions on the buckling behavior of FG-GRC shells*. Mechanics of Advanced Materials, 92, 190012.
10. Kumar, A., et al. (2021). *A study on imperfection sensitivity in FG graphene composite plates*. Journal of Composite Structures, 263, 113723.
11. Zhao, Y. (2019). *Thermal buckling of nano-reinforced composite plates*. Composite Interfaces, 26(2), 145–158.

12. Patil, R., & Mehta, N. (2020). *Post-buckling performance of FG-GRC panels under mechanical loads*. Applied Mechanics Reviews, 72(3), 030803.
13. Tang, H., & Li, X. (2021). *Critical buckling evaluation of thermally loaded GRC beams*. Composite Materials, 54, 3911–3924.
14. Sun, W. (2022). *Mesh convergence studies in high-fidelity composite simulations*. International Journal of Structural Integrity, 13(1), 1–15.
15. Wang, Y., & Feng, K. (2016). *Unified solution for FG-GRC circular plates under axisymmetric buckling*. Engineering Structures, 120, 90–101.
16. Shen, H. (2018). *Thermo-mechanical responses of FG-GRC structures: A review*. Journal of Composite Materials, 52(5), 703–720.
17. El-Mansy, M., & Al-Amri, S. (2019). *Graphene composites under thermal and dynamic loads*. Advances in Composite Materials, 28, 161–172.
18. Khandoker, S., & Rahman, H. (2020). *Postbuckling behavior of FG nanocomposite panels using ABAQUS*. Finite Elements in Analysis and Design, 180, 103379.
19. Narayan, R., et al. (2021). *Critical load patterns in functionally graded composites*. Procedia Structural Integrity, 31, 121–128.
20. Chen, J., & Liu, T. (2022). *Multi-scale modeling of FG-GRC using UMAT in ABAQUS*. International Journal of Solids and Structures, 250, 111718.